

Unit I: Motion in One and Two Dimensions

Distance, displacement, velocity, acceleration, equations of motion, projectile motion, and motion graphs

Learning Goals

- Distinguish between distance and displacement.
- Understand speed, velocity, and acceleration as descriptions of motion.
- Apply equations of motion for uniformly accelerated motion.
- Analyze motion in one dimension using signs, graphs, and formulas.
- Resolve two-dimensional motion into horizontal and vertical components.
- Study projectile motion using independent x and y directions.
- Interpret displacement-time, velocity-time, and acceleration-time graphs.

1. Introduction to Motion

Mechanics is the branch of physics that studies motion and the causes of motion. Kinematics describes motion without discussing forces. In this unit, we describe how position changes with time using distance, displacement, velocity, and acceleration.

Motion can occur in one dimension, such as a car moving along a straight road, or in two dimensions, such as a ball thrown through the air. The same basic ideas apply, but two-dimensional motion is handled by breaking the motion into perpendicular components.

2. Distance and Displacement

Distance is the total path length travelled by an object. It is a scalar quantity, so it has magnitude only and no direction. Distance is always zero or positive.

Displacement is the change in position from the initial point to the final point. It is a vector quantity, so it has both magnitude and direction. Displacement can be positive, negative, or zero depending on the chosen coordinate direction.

Distance = total path length

Displacement = final position - initial position

Delta x = x_f - x_i

Quantity	Meaning
Distance	Total length of path covered; scalar.
Displacement	Shortest directed change from initial to final position; vector.
Example	Walking 3 m east and 3 m west gives distance 6 m but displacement 0 m.

- Distance depends on the actual path taken.
- Displacement depends only on initial and final positions.
- For straight-line motion in one direction, distance and displacement magnitudes may be equal.

3. Speed and Velocity

Speed tells how fast an object moves. It is a scalar. Velocity tells how fast position changes in a specified direction. It is a vector. Two objects can have the same speed but different velocities if they move in different directions.

$$\text{Average speed} = \text{total distance} / \text{total time}$$

$$\text{Average velocity} = \text{displacement} / \text{time interval} = \Delta x / \Delta t$$

Instantaneous velocity is the velocity at a particular moment. In calculus language, it is the derivative of position with respect to time. In graph language, it is the slope of the displacement-time graph at that instant.

$$\text{Instantaneous velocity: } v = dx/dt$$

- If velocity is positive, the object moves in the positive direction.
- If velocity is negative, the object moves in the negative direction.
- If velocity is zero, the object is momentarily at rest.

4. Acceleration

Acceleration is the rate of change of velocity with time. It is a vector quantity. An object accelerates when its speed changes, its direction changes, or both.

$$\text{Average acceleration} = \text{change in velocity} / \text{time interval} = \Delta v / \Delta t$$

$$\text{Instantaneous acceleration: } a = dv/dt = d^2x/dt^2$$

- Positive acceleration means acceleration points in the positive direction.
- Negative acceleration means acceleration points in the negative direction.
- Acceleration opposite to velocity reduces speed.
- Acceleration in the same direction as velocity increases speed.

In freely falling motion near the Earth's surface, acceleration due to gravity is approximately 9.8 m/s^2 downward. If upward is chosen as positive, then $a = -g = -9.8 \text{ m/s}^2$.

5. Equations of Motion

For motion with constant acceleration in a straight line, a set of standard equations connects displacement, velocity, acceleration, and time. These equations are also called kinematic equations.

$$v = u + at$$

$$s = ut + (1/2)at^2$$

$$v^2 = u^2 + 2as$$

$$s = [(u + v)/2] t$$

Symbol	Meaning
u	Initial velocity.
v	Final velocity.
a	Constant acceleration.
t	Time interval.
s	Displacement during the time interval.

- Use these equations only when acceleration is constant.
- Choose a positive direction before substituting values.
- Use signs carefully for velocity, displacement, and acceleration.
- For vertical motion, gravity acts downward, so its sign depends on the chosen axis.

6. Motion in One Dimension

One-dimensional motion occurs along a straight line. Examples include an elevator moving up and down, a train on a straight track, or a stone dropped from rest. The direction of motion can be represented using positive and negative signs.

Uniform Motion

Uniform motion means velocity is constant. The acceleration is zero, and equal displacements are covered in equal time intervals.

$$\text{For uniform motion: } x = x_0 + vt$$

Uniformly Accelerated Motion

Uniformly accelerated motion means acceleration is constant. Velocity changes by equal amounts in equal time intervals. The equations of motion are used for this case.

- A car speeding up uniformly has constant positive acceleration.
- A ball thrown upward has constant downward acceleration due to gravity.

- At the highest point of vertical motion, velocity is zero but acceleration is still downward.

7. Motion in Two Dimensions

Two-dimensional motion occurs in a plane. Position, velocity, and acceleration are represented using components along two perpendicular axes, usually x and y. A vector can be resolved into horizontal and vertical components.

Position vector: $\mathbf{r} = x \mathbf{i} + y \mathbf{j}$

Velocity: $\mathbf{v} = v_x \mathbf{i} + v_y \mathbf{j}$

Acceleration: $\mathbf{a} =$

Magnitude: $|\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$

Direction: $\tan(\theta) = v_y / v_x$

The key idea is independence of perpendicular components. Horizontal motion and vertical motion can be analyzed separately, then combined to describe the full motion.

- Horizontal and vertical components are perpendicular.
- A change in one component does not automatically mean the other component changes.
- Two-dimensional problems are often solved by writing separate x-direction and y-direction equations.

8. Projectile Motion

Projectile motion is the motion of an object thrown or projected into the air under the influence of gravity alone, neglecting air resistance. The path of a projectile is a parabola.

Projectile motion is analyzed by resolving initial velocity into horizontal and vertical components. The horizontal component remains constant if air resistance is ignored. The vertical component changes because gravity acts downward.

$u_x = u \cos(\theta)$

$u_y = u \sin(\theta)$

Horizontal: $x = u_x t$

Vertical: $y = u_y t - (1/2)gt^2$

$v_y = u_y - gt$

$v_x = u_x$

Important Results for Same Launch and Landing Level

Time of flight: $T = 2u \sin(\theta) / g$

Maximum height: $H = u^2 \sin^2(\theta) / (2g)$

Range: $R = u^2 \sin(2\theta) / g$

- Maximum range occurs at $\theta = 45$ degrees when launch and landing heights are equal.
- Angles θ and $90 - \theta$ degrees give the same range for the same speed.
- At the highest point, vertical velocity is zero but horizontal velocity remains unchanged.
- Acceleration is always g downward throughout the flight.

9. Motion Graphs

Graphs provide a visual way to understand motion. The three most important graphs are displacement-time, velocity-time, and acceleration-time graphs.

Displacement-Time Graph

- The slope of a displacement-time graph gives velocity.
- A straight line means constant velocity.
- A horizontal line means the object is at rest.
- A curved graph means velocity is changing.

$$\text{Slope of } x\text{-}t \text{ graph} = \Delta x / \Delta t = \text{velocity}$$

Velocity-Time Graph

- The slope of a velocity-time graph gives acceleration.
- The area under a velocity-time graph gives displacement.
- A horizontal line means constant velocity and zero acceleration.
- A straight sloping line means constant acceleration.

$$\text{Slope of } v\text{-}t \text{ graph} = \Delta v / \Delta t = \text{acceleration}$$

$$\text{Area under } v\text{-}t \text{ graph} = \text{displacement}$$

Acceleration-Time Graph

- The area under an acceleration-time graph gives change in velocity.
- A horizontal acceleration-time graph means constant acceleration.
- If acceleration is zero, velocity remains constant.

$$\text{Area under } a\text{-}t \text{ graph} = \text{change in velocity}$$

10. Worked Examples

Example 1: Distance and Displacement

A student walks 5 m east and then 3 m west. Distance = $5 + 3 = 8$ m. Displacement = $5 - 3 = 2$ m east.

Example 2: Uniform Acceleration

A car starts from rest and accelerates at 2 m/s^2 for 5 s. Using $v = u + at$, $v = 0 + 2(5) = 10 \text{ m/s}$. Using $s = ut + (1/2)at^2$, $s = 0 + (1/2)(2)(25) = 25 \text{ m}$.

Example 3: Projectile Components

A ball is thrown with speed u at angle θ . Its horizontal component is $u \cos(\theta)$, and its vertical component is $u \sin(\theta)$. The horizontal part controls range, while the vertical part controls time of flight and maximum height.

11. Common Mistakes

- Confusing distance with displacement.

- Treating speed and velocity as the same quantity.
- Forgetting that acceleration can exist even when velocity is momentarily zero.
- Using equations of motion when acceleration is not constant.
- Ignoring signs when solving one-dimensional motion problems.
- Assuming horizontal acceleration exists in projectile motion without air resistance.
- Forgetting that the area under a velocity-time graph gives displacement, not distance in all cases.

Quick Revision Sheet

- Distance is total path length; displacement is directed change in position.
- Speed is scalar; velocity is vector.
- Acceleration is the rate of change of velocity.
- Equations of motion apply only for constant acceleration.
- In vertical motion near Earth, acceleration due to gravity is downward.
- Two-dimensional motion is solved by resolving vectors into x and y components.
- Projectile motion has constant horizontal velocity and constant downward acceleration.
- The path of a projectile is parabolic when air resistance is ignored.
- Slope of displacement-time graph gives velocity.
- Slope of velocity-time graph gives acceleration.
- Area under velocity-time graph gives displacement.
- Area under acceleration-time graph gives change in velocity.

Practice Questions

- Define distance and displacement with one example where they are different.
- A body moves from $x = 2 \text{ m}$ to $x = 9 \text{ m}$. Find its displacement.
- A car travels 120 m in 10 s. Find its average speed.
- A body changes velocity from 5 m/s to 25 m/s in 4 s. Find acceleration.
- A body starts from rest with acceleration 3 m/s^2 for 6 s. Find final velocity and displacement.
- A stone is thrown vertically upward. What are velocity and acceleration at the highest point?
- Resolve a velocity of 20 m/s at 30 degrees into horizontal and vertical components.
- State the formulas for time of flight, maximum height, and range of a projectile.
- Why is horizontal velocity constant in ideal projectile motion?
- What does the slope of a displacement-time graph represent?
- What does the area under a velocity-time graph represent?
- Draw and explain the velocity-time graph for uniformly accelerated motion.